

CHATBOT MANAGEMENT SYSTEM USING C++

A Console-Based Application Using Object-Oriented Programming

Presented to

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In Partial Fulfillment

of the Requirements for the Subject

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System Architecture Overview

The Chatbot Management System follows a simple console-based architecture developed entirely in one `main.cpp` file. The architecture focuses on user interaction, input processing, logical decision-making, and chatbot response generation using Object-Oriented Programming principles.

Architecture Components

1. User Interface Layer

This layer handles the interaction between the user and the chatbot through the console terminal.

Functions:

- Accepts user input
- Displays chatbot responses
- Shows system messages

2. Processing Layer

This layer contains the chatbot logic.

Functions:

- Processes user input
- Uses conditional statements
- Matches keywords and commands
- Determines chatbot responses

3. OOP Structure Layer

This layer contains the class and object implementation.

Functions:

- Organizes chatbot behavior
- Encapsulates functions inside a class
- Improves code readability and maintainability

4. Control Flow Layer

This layer controls the looping mechanism of the program.

Functions:

- Maintains continuous interaction
- Repeats chatbot execution using loops
- Handles exit conditions safely

System Flow

1. Program starts
2. Chatbot object is initialized
3. User enters message
4. System processes input
5. Chatbot generates response
6. Response is displayed
7. Loop repeats until user enters exit command
8. Program terminates

Architectural Design Used

The project uses a simple linear and object-oriented architecture because the system is console-based and beginner-friendly. The architecture prioritizes:

- Simplicity
- Readability
- Maintainability
- Logical program flow
- Efficient debugging

This design makes the system easy to understand while still demonstrating proper programming structure and Object-Oriented Programming implementation.

Introduction

The Chatbot Management System is a console-based application developed using C++ and Object-Oriented Programming (OOP). The system allows users to interact with a chatbot through text input and receive automated responses based on predefined conditions. The project was designed to demonstrate programming fundamentals such as classes, objects, functions, loops, conditional statements, and user input processing. All functionalities of the system are implemented inside one file named `main.cpp`

Objectives

The objectives of the project are:

- To create a simple chatbot system using C++

- To apply Object-Oriented Programming concepts
- To implement user interaction using loops and conditions
- To demonstrate proper program flow and logic
- To organize the system in a structured and readable way

Scope of the System

The system is limited to:

- Console-based interaction
- Predefined chatbot responses
- Basic keyword matching
- Simple user input processing

The system does not include:

- Artificial intelligence
- Database integration
- Internet connection
- Voice recognition
- Graphical User Interface (GUI)

SDG Connection

The project supports SDG 4: Quality Education.

The chatbot system helps students understand:

- Programming concepts

- Object-Oriented Programming
- Problem-solving skills
- Logical thinking
- Technical software development

Programming Language and Tools

Programming Language

- C++

Tools Used

- VS Code / Dev-C++
- GCC Compiler
- Console Terminal

Main File

- `Main.cpp`
- `ChatbotManager.cpp`
- `ChatbotManager.h`
- `Inquiry.h`
- `Inquiry.cpp`

System Features

The chatbot system can:

- Accept user input
- Display chatbot responses
- Continue conversation using loops
- Exit the program safely
- Process simple commands and keywords

Object-Oriented Programming Concepts

Class

A class is used as the blueprint of the chatbot system.

Object

An object is created from the chatbot class to access chatbot functions.

Encapsulation

Functions are grouped inside the class to organize the code properly.

Functions

Functions are used to process user input and generate responses.

Program Flow

1. Start the program
2. Display welcome message
3. Ask for user input
4. Process the input
5. Display chatbot response
6. Repeat the process
7. End the program when user types `exit`

Technical Implementation

The system uses:

- `getline()` for user input
- `if-else` statements for decision making
- `while` loop for continuous interaction
- Functions for response handling
- String comparison for command processing

Sample Code Structure

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class Chatbot {
```

```
public:
```

```
    void respond(string input) {
```



```

if(input == "hello") {
    cout << "Hello!" << endl;
}
}
};

```

Sample Output

```

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0. Exit
Enter choice:
Invalid input.

===== She4Tech SDG Inquiry System =====
1. Add Inquiry
2. View Inquiries
3. Search Inquiry
0. Exit
Enter choice: 1
Enter ID: 1
Enter Name: Angela
Enter Inquiry: Food Assistance
Inquiry added successfully.

===== She4Tech SDG Inquiry System =====
1. Add Inquiry
2. View Inquiries
3. Search Inquiry
0. Exit
Enter choice: 2

ID: 0
Name:
Inquiry:
=====

ID: 1
Name: Angela
Inquiry: Food Assistance
=====

```

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===== She4Tech SDG Inquiry System =====

1. Add Inquiry

2. View Inquiries

3. Search Inquiry

0. Exit

Enter choice: 1

Enter ID: 2

Enter Name: Maria

Enter Inquiry: Scholarship

Inquiry added successfully.

===== She4Tech SDG Inquiry System =====

1. Add Inquiry

2. View Inquiries

3. Search Inquiry

0. Exit

Enter choice: 2

ID: 0

Name:

Inquiry:

ID: 1

Name: Angela

Inquiry: Food Assistance

ID: 2

Name: Maria

Inquiry: Scholarship

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===== She4Tech SDG Inquiry System =====

1. Add Inquiry

2. View Inquiries

3. Search Inquiry

0. Exit

Enter choice: 1

Enter ID: 3

Enter Name: Joac

Enter Inquiry: Water Sanitation Concern

Inquiry added successfully.

===== She4Tech SDG Inquiry System =====

1. Add Inquiry

2. View Inquiries

3. Search Inquiry

0. Exit

Enter choice: 2

ID: 0

Name:

Inquiry:

ID: 1

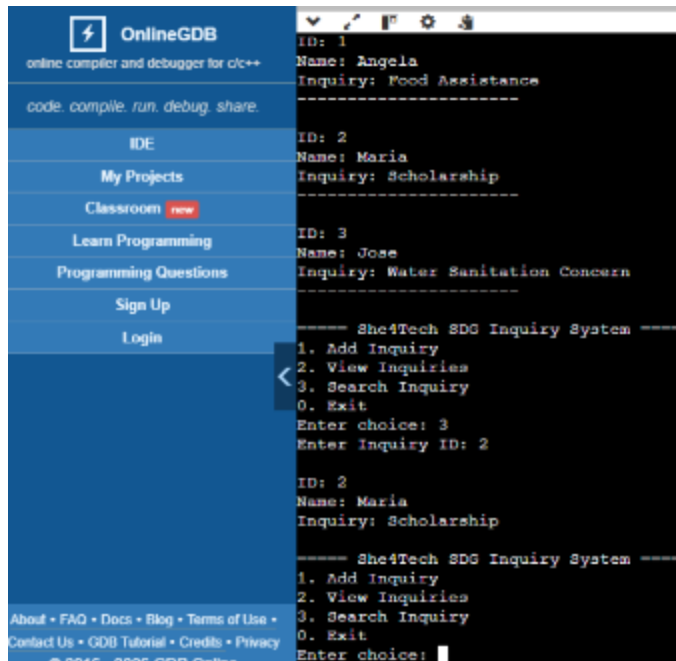
Name: Angela

Inquiry: Food Assistance

ID: 2

Name: Maria

Inquiry: Scholarship



Problems Encountered

During development, the following problems were encountered:

- Missing header file error
- Looping issues
- Input handling problems
- String comparison mistakes

Solutions Applied

The following solutions were implemented:

- Combined the entire system into one file
- Fixed loop conditions
- Improved string handling

- Simplified program structure

Conclusion

The Chatbot Management System successfully demonstrated the use of Object-Oriented Programming in C++. The project applied important programming concepts such as classes, objects, functions, loops, and conditional statements. Although the system is simple, it effectively shows proper coding structure, logical program flow, and technical implementation. The project also supports SDG 4: Quality Education by promoting programming knowledge and technical learning.

Recommendations

For future improvements, the system may include:

- Artificial intelligence features
- Database integration
- Graphical User Interface (GUI)
- Voice recognition
- Advanced chatbot responses
- Online connectivity

References

1. Stroustrup, B. (2013). *Programming: Principles and Practice Using C++*. Addison-Wesley.
2. Deitel, P., & Deitel, H. (2017). *C++ How to Program*. Pearson Education.

3. Object-Oriented Programming Concepts in C++. Retrieved from
<https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/>
4. C++ Documentation. Retrieved from <https://cplusplus.com/>
5. Sustainable Development Goal 4: Quality Education. United Nations. Retrieved from
<https://sdgs.un.org/goals/goal4>